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A JORNADA DO HERÓI | 30 ANOS DE INOVAÇÃO



Criação de jogos em Delphi

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Descubra a jornada de criação de jogos em Delphi, desde a vitória no concurso [radprogrammer-challenge1](#) com Minesweeper até uma releitura do clássico Arkanoid.

Nesta palestra, compartilho os desafios e soluções no desenvolvimento desses jogos, com destaque para técnicas de animação de personagens usando bitmaps e a biblioteca SKIA

Do Conceito ao Jogo: A Jornada do Desenvolvimento em Delphi

Desenvolver jogos é transformar ideias em experiências interativas.

Nesta seção, exploramos os passos essenciais do processo:

- Brainstorming de conceitos
- Planejamento da mecânica
- Design de assets
- Trilha sonora
- Programação

Case 1 – MINESWEEPER

Vencedor do concurso radprogrammer-challenge1

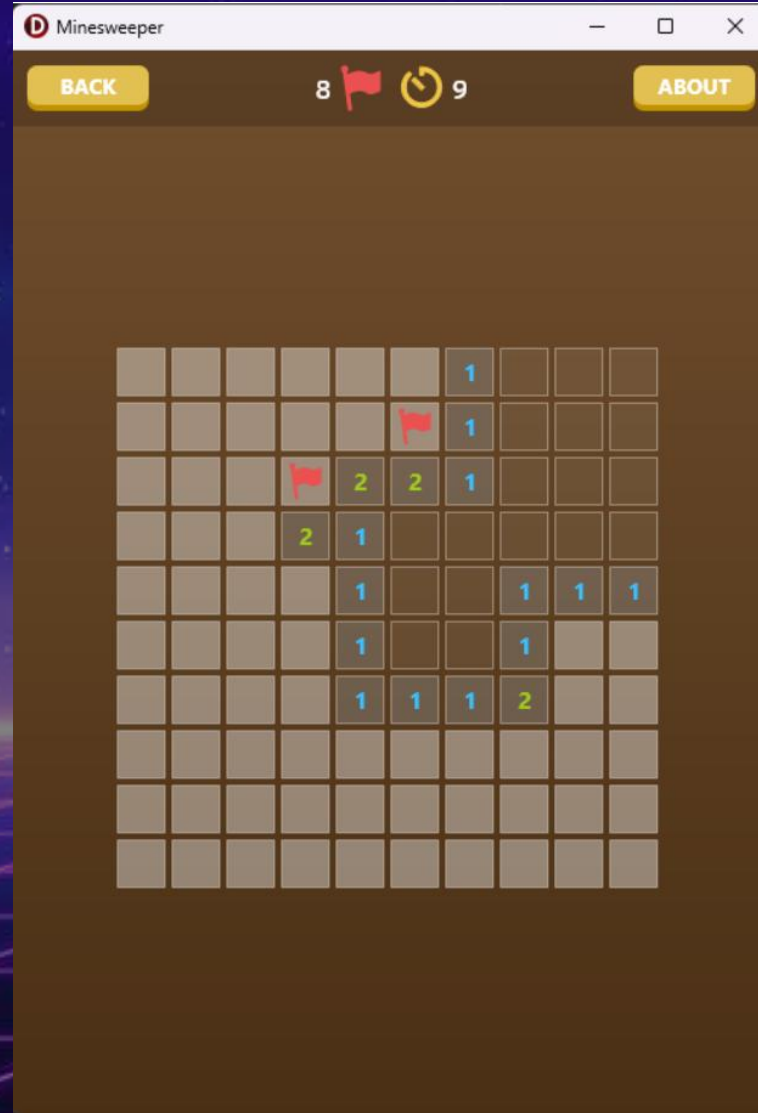
<https://play.google.com/store/apps/details?id=br.com.imperium.MineSweeper>

Brainstorming de conceitos

- Ambientação (cenário, narrativas, estilo visual)
- Gênero (quebra cabeça, ação, RPG)
- Inspirações e/ou referências (outros jogos, filmes)
- Diferenciais (em que eu quero que ele se destaque)



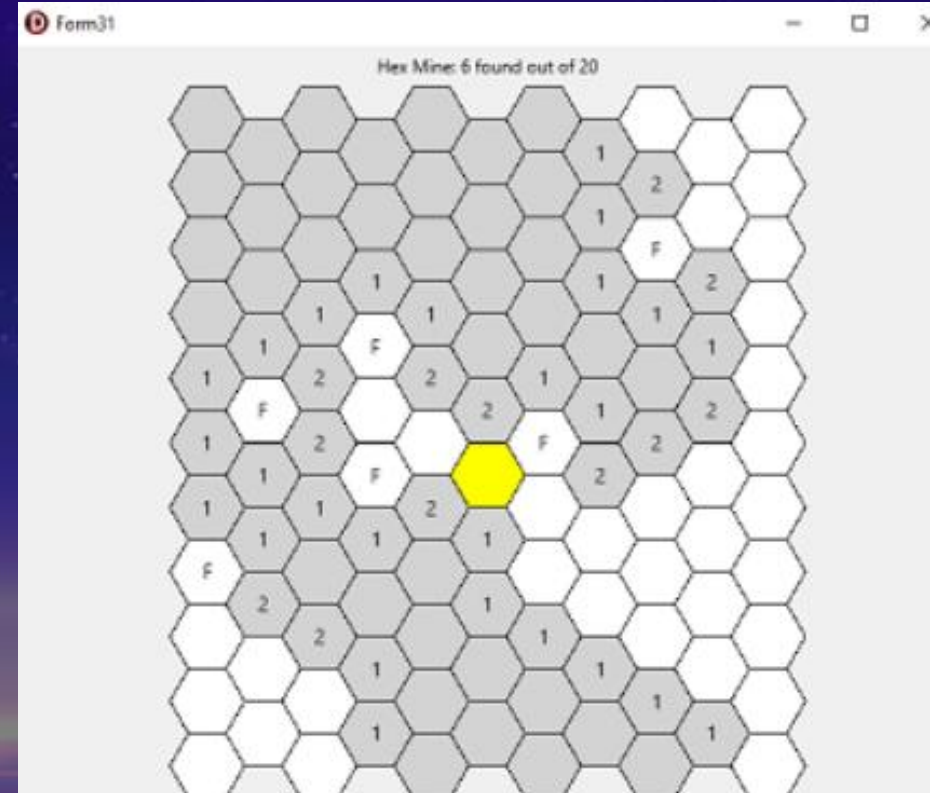
Outros competidores



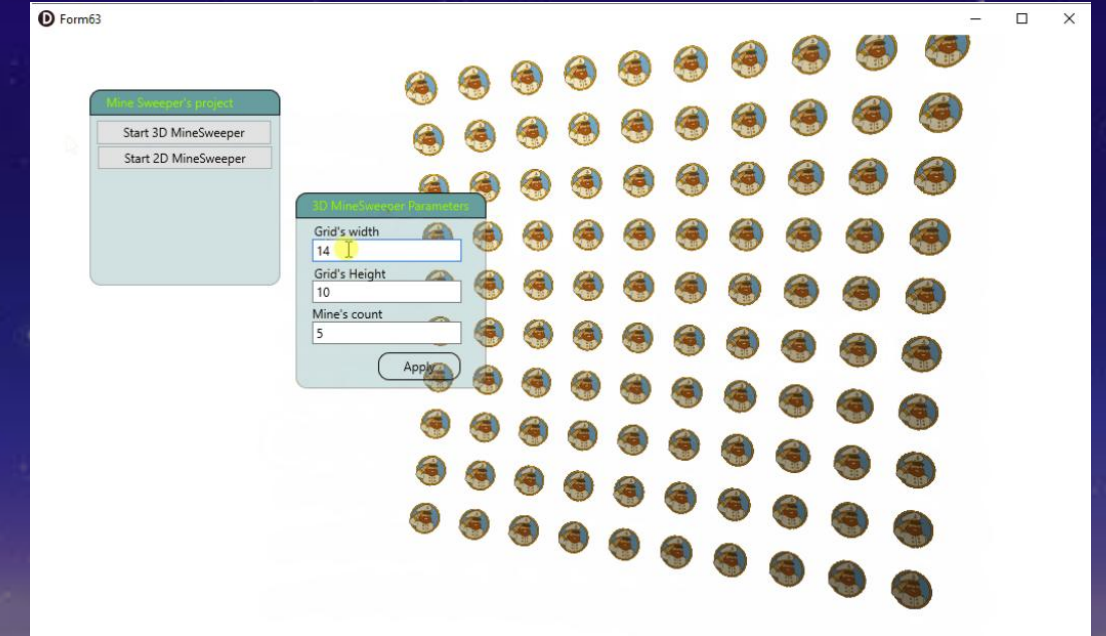
<https://github.com/HemuIGM>



<https://github.com/abritolda1972>



<https://github.com/jimmckeeth>



<https://github.com/VincentGsell>

Gameplay

Animações:

- Fade no contorno dos campos
- Passagem do P38
- Zoom in/out no campo de jogo
- Seleção do tamanho do campo e da dificuldade



Códigos Fade no contorno dos campos

```
procedure MouseUp(Button: TMouseButton; Shift: TShiftState; X, Y: Single); Override;
50 public
  Constructor Create(aOwner : TComponent); Override;
  Procedure ShowIn(aForm : TFMXObject);
  Procedure CentralizeField(aField : Integer);
  Procedure CentralizeMap;
  Procedure HideMessage;
  Procedure ShowMessage;
  Property AniOpacity : Single Read FAniOpacity Write SetAniOpacity;
  Property SelectedField : Integer Read FSelectedField Write SetSelectedField;
  Property RectImage : TRectF Read FRectImage Write SetRectImage;
  Property ImageRatio : Single Read FImageRatio Write SetImageRatio;
60 Property ImageOffsetX : Single Read FImageOffsetX Write SetImageOffsetX;
  Property ImageOffsetY : Single Read FImageOffsetY Write SetImageOffsetY;
  Property FlashFields : Boolean Read FFlashFields Write SetFlashFields;
end;
```

```
300
301 procedure TStartScreen.SetAniOpacity(const Value: Single);
begin
  FAniOpacity := Value;
  Repaint;
end;
```

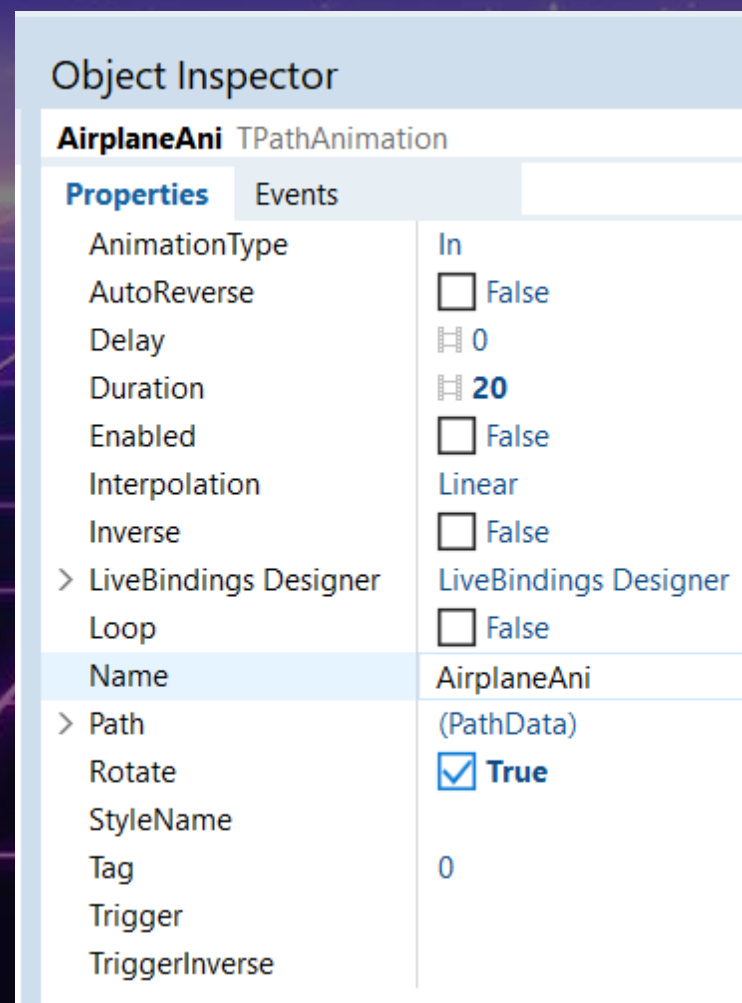
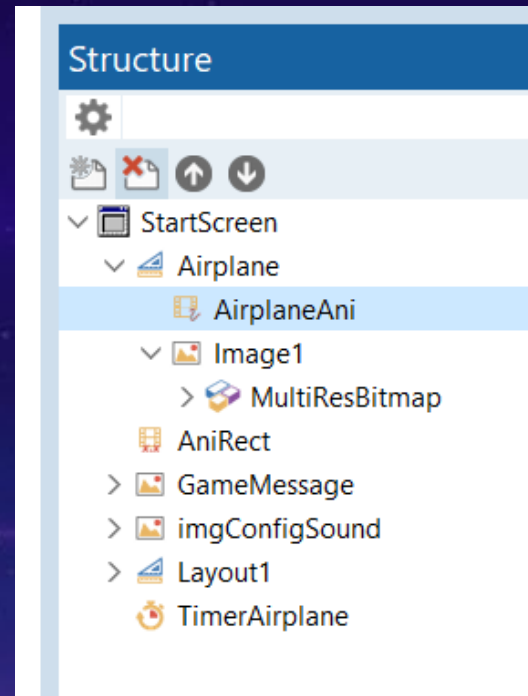
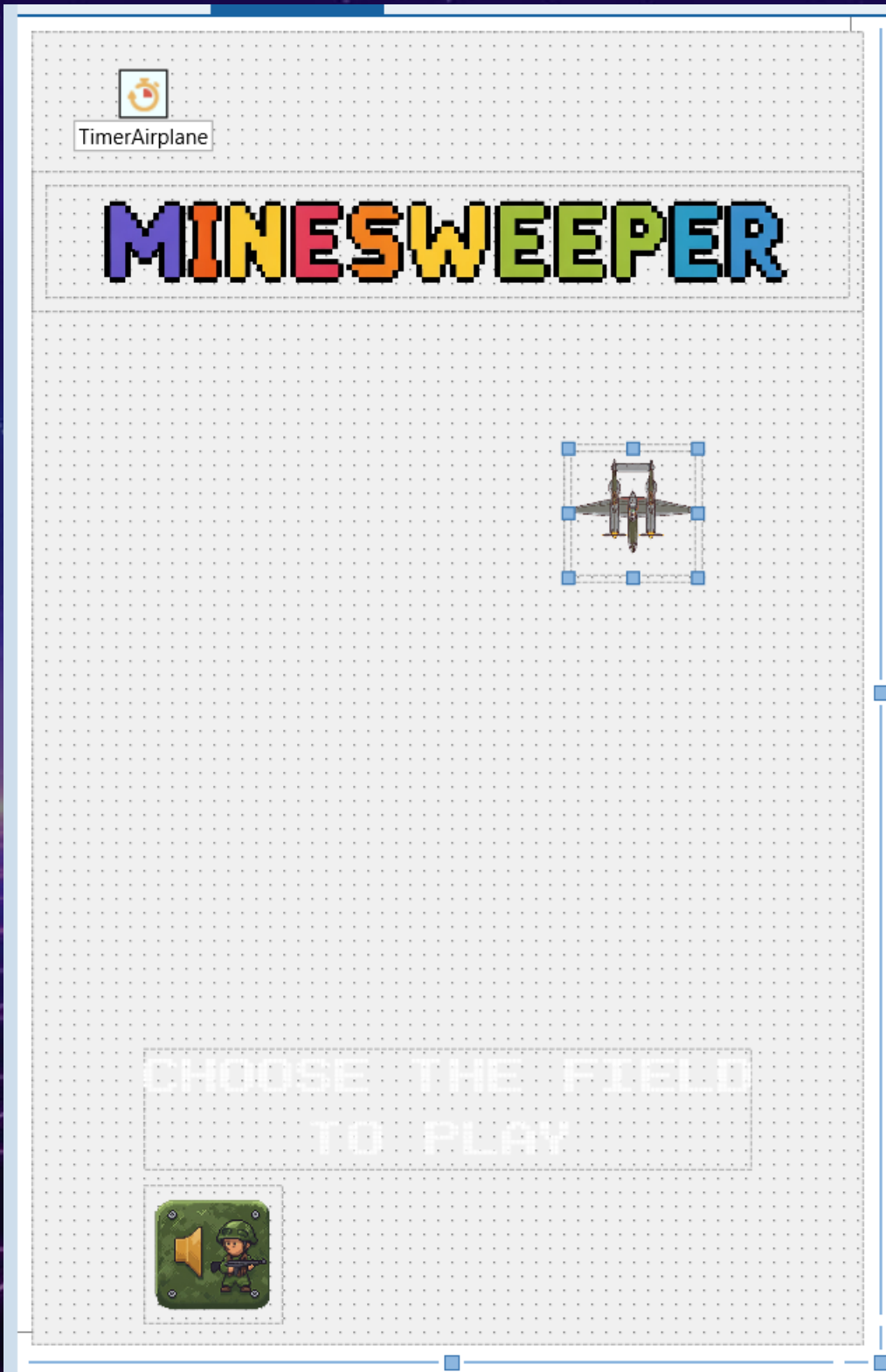
```
FImageOffsetY := 0;
SetLength(FFieldsRect, 7);
FFieldsRect[0] := TRectF.Create( 286, 745, 505, 872);
FFieldsRect[1] := TRectF.Create( 620, 638, 830, 783);
FFieldsRect[2] := TRectF.Create(1048, 806, 1265, 948);
FFieldsRect[3] := TRectF.Create(1480, 489, 1700, 633);
FFieldsRect[4] := TRectF.Create( 423, 1195, 657, 1336);
FFieldsRect[5] := TRectF.Create(1323, 1203, 1539, 1344);
FFieldsRect[6] := TRectF.Create( 548, 1434, 749, 1578);
170 AniRect.PropertyName := 'AniOpacity';
  AniRect.Enabled := True;
  GameMessage.Opacity := 0;
```

```
procedure TStartScreen.Paint;
begin
  FAniOpacity := Value;
  Repaint;
end;

var
  i : Integer;
  LRect : TRectF;

280 begin
  inherited;
  Canvas.Fill.Color := $ff2e5d70;
  Canvas.Fill.Kind := TBrushKind.Solid;
  Canvas.FillRect(BoundsRect, 1);
  FRectImage.SetLocation((Width-FRectImage.Width)/2+FImageOffsetX, (Height-FRectImage.Height)/2+FImageOffsetY);
  Canvas.DrawBitmap(FBackground, FBackground.BoundsF, FRectImage, 1);
  Canvas.Stroke.Kind := TBrushKind.Solid;
  Canvas.Stroke.Thickness := 3;
  For i := 0 To 6 Do
290   Begin
     LRect := ProportionalRect(FFieldsRect[i]);
     LRect.Offset(FRectImage.Left, FRectImage.Top);
     If i = FSelectedField Then
        Canvas.Stroke.Color := TAlphaColors.Greenyellow
     Else
        Canvas.Stroke.Color := TAlphaColors.White;
     Canvas.DrawRect(LRect, FAniOpacity);
   End;
  End;
```

Códigos Passagem do P38



```

- InnerRect;
- Airplane.Position.x := 0;
- Airplane.Position.y := -90;
- Setlength(FRotas, 4);
- FRota := 0;
- FRotas[3] := 'M 60.957 8.282 '+
.           'C 72.418 84.734 272.376 63.067 348.248 98.651 '+
.           'C 413.606 129.304 358.481 221.54 346.239 233.997 '+
130 .           'C 259.838 321.91 120.175 290.655 127.294 224.273 '+
.           'C 136.1 142.16 200.272 141.101 266.451 150.604 '+
.           'C 300.593 155.507 454.929 231.646 340.258 297.847 '+
.           'C 21.421 481.916 296.289 487.42 319.364 495.942';
- FRotas[2] := 'M 416.763 25.178 '+
.           'C 438.097 167.488 372.211 206.614 346.239 233.997 '+
.           'C 264.331 320.355 211.595 270.183 196.534 237.525 '+
.           'C 171.296 182.799 210.181 153.164 266.451 150.604 '+
.           'C 300.908 149.036 408.125 184.154 340.258 297.847 '+
.           'C 280.664 397.682 9.286 479.081 30.147 488.322';
140 FRotas[1] := 'M 163.657 2.318 '+
.           'C 175.118 78.77 182.816 107.327 191.216 132.443 '+
.           'C 205.786 176.01 266.606 231.065 204.115 296.28 '+
.           'C 118.834 385.28 23.266 288.781 29.232 222.285 '+
.           'C 33.883 170.444 72.854 145.796 122.671 146.297 '+
.           'C 194.999 147.024 292.858 174.519 265.386 324.019 '+
.           'C 255.325 378.773 87.082 404.231 141.461 498.593';
- FRotas[0] := 'M 216.995 1.655 '+
.           'C 94.283 47.297 30.119 108.777 23.583 148.675 '+
.           'C 6.534 252.742 163.502 99.708 208.091 182.979 '+
150 .           'C 259.428 278.852 -15.888 347.317 29.232 222.285 '+
.           'C 46.9 173.326 58.853 118.804 103.456 140.997 '+
.           'C 137.619 157.995 201.021 156.45 172.562 235.986 '+
.           'C 165.737 255.059 134.783 251.772 93.305 254.993 '+
.           'C 58.549 257.692 25.76 320.222 48.059 346.215 '+
.           'C 88.4 393.241 171.929 397.972 141.461 498.593';

```


Códigos Zoom in/out no campo de jogo - 1

```
- SetLength(FFieldsRect, 7);  
- FFieldsRect[0] := TRectF.Create( 286, 745, 505, 872);  
- FFieldsRect[1] := TRectF.Create( 620, 638, 830, 783);  
- FFieldsRect[2] := TRectF.Create(1048, 806, 1265, 948);  
- FFieldsRect[3] := TRectF.Create(1480, 489, 1700, 633);  
- FFieldsRect[4] := TRectF.Create( 423, 1195, 657, 1336);  
- FFieldsRect[5] := TRectF.Create(1323, 1203, 1539, 1344);  
- FFieldsRect[6] := TRectF.Create( 548, 1434, 749, 1578);
```

```
- procedure TStartScreen.MouseUp(Button: TMouseButton; Shift: TShiftState; X, Y: Single);  
- Var  
- I : Integer;  
- LRect : TRectF;  
- begin  
- inherited;  
260 Media.Click2;  
- If FieldView.Parent <> Nil Then Exit;  
- For i := 0 To 6 Do  
- Begin  
- LRect := ProportionalRect(FFieldsRect[i]);  
- LRect.Offset(FRectImage.Left, FRectImage.Top);  
- If LRect.Contains(TPointF.Create(X, y)) Then  
- Begin  
- FSelectedField := i;  
- FrameInstructions.SetField(i+1);  
270 FrameInstructions.ShowIn(Self);  
- Break;  
- End;  
- End;  
- end;
```

```
- procedure TStartScreen.Paint;  
- Var  
- i : Integer;  
- LRect : TRectF;  
280 begin  
- inherited;  
- Canvas.Fill.Color := $ff2e5d70;  
- Canvas.Fill.Kind := TBrushKind.Solid;  
- Canvas.FillRect(BoundsRect, 1);  
- FRectImage.SetLocation((Width-FRectImage.Width)/2+FImageOffsetX, (Height-FRectImage.Height)/2+FImageOffsetY);  
- Canvas.DrawBitmap(FBackground, FBackground.BoundsF, FRectImage, 1);  
- Canvas.Stroke.Kind := TBrushKind.Solid;  
- Canvas.Stroke.Thickness := 3;  
- For i := 0 To 6 Do  
290 Begin  
- LRect := ProportionalRect(FFieldsRect[i]);  
- LRect.Offset(FRectImage.Left, FRectImage.Top);  
- If i = FSelectedField Then  
- Canvas.Stroke.Color := TAlphaColors.Greenyellow  
- Else  
- Canvas.Stroke.Color := TAlphaColors.White;  
- Canvas.DrawRect(LRect, FAniOpacity);  
- End;  
300 end;
```

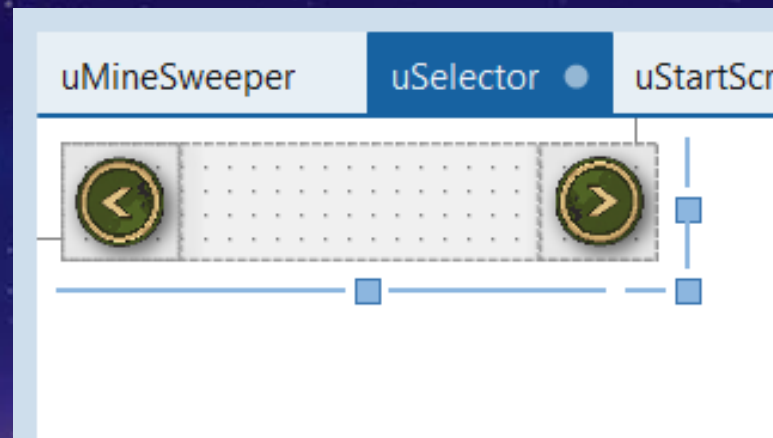

Códigos

Zoom in/out no campo de jogo - 2

```
- procedure TStartScreen.CentralizeField(aField: Integer);  
- Var  
-     LRatio    : Single;  
-     LRectOff  : TRectF;  
-     LRealRat  : Single;  
-     LOffX     : Single;  
-     LOffY     : Single;  
100 begin  
-     LRectOff := ProportionalRect(FFieldsRect[aField]);  
-     LRatio   := Width / LRectOff.Height;  
-     LRealRat := LRatio * FStartImageRatio;  
-     LOffX    := (FFieldsRect[aField].Left * LRealRat)+(Width - (FBackground.Width * LRealRat))/2;  
-     LOffX    := LOffX + ((FFieldsRect[aField].Width*LRealRat)-(FFieldsRect[aField].Height*LRealRat)) / 2;  
-     LOffY    := (FFieldsRect[aField].Top * LRealRat)+(Height - (FBackground.Height * LRealRat))/2;  
-     LOffY    := LOffY - (Height - FFieldsRect[aField].Height*LRealRat)/2;  
-     TAnimator.AnimateFloat(Self, 'ImageRatio',    LRatio, 1.5, TAnimationType.InOut, TInterpolationType.Exponential);  
-     TAnimator.AnimateFloat(Self, 'ImageOffsetX', -LOffX, 1.5, TAnimationType.InOut, TInterpolationType.Exponential);  
110 TAnimator.AnimateFloat(Self, 'ImageOffsetY', -LOffY, 1.5, TAnimationType.InOut, TInterpolationType.Exponential);  
- end;  
  
- Procedure TStartScreen.CentralizeMap;  
- Begin  
-     TAnimator.AnimateFloat(Self, 'ImageRatio',    1, 1.5, TAnimationType.InOut, TInterpolationType.Exponential);  
-     TAnimator.AnimateFloat(Self, 'ImageOffsetX', 0, 1.5, TAnimationType.InOut, TInterpolationType.Exponential);  
-     TAnimator.AnimateFloat(Self, 'ImageOffsetY', 0, 1.5, TAnimationType.InOut, TInterpolationType.Exponential);  
- End;
```

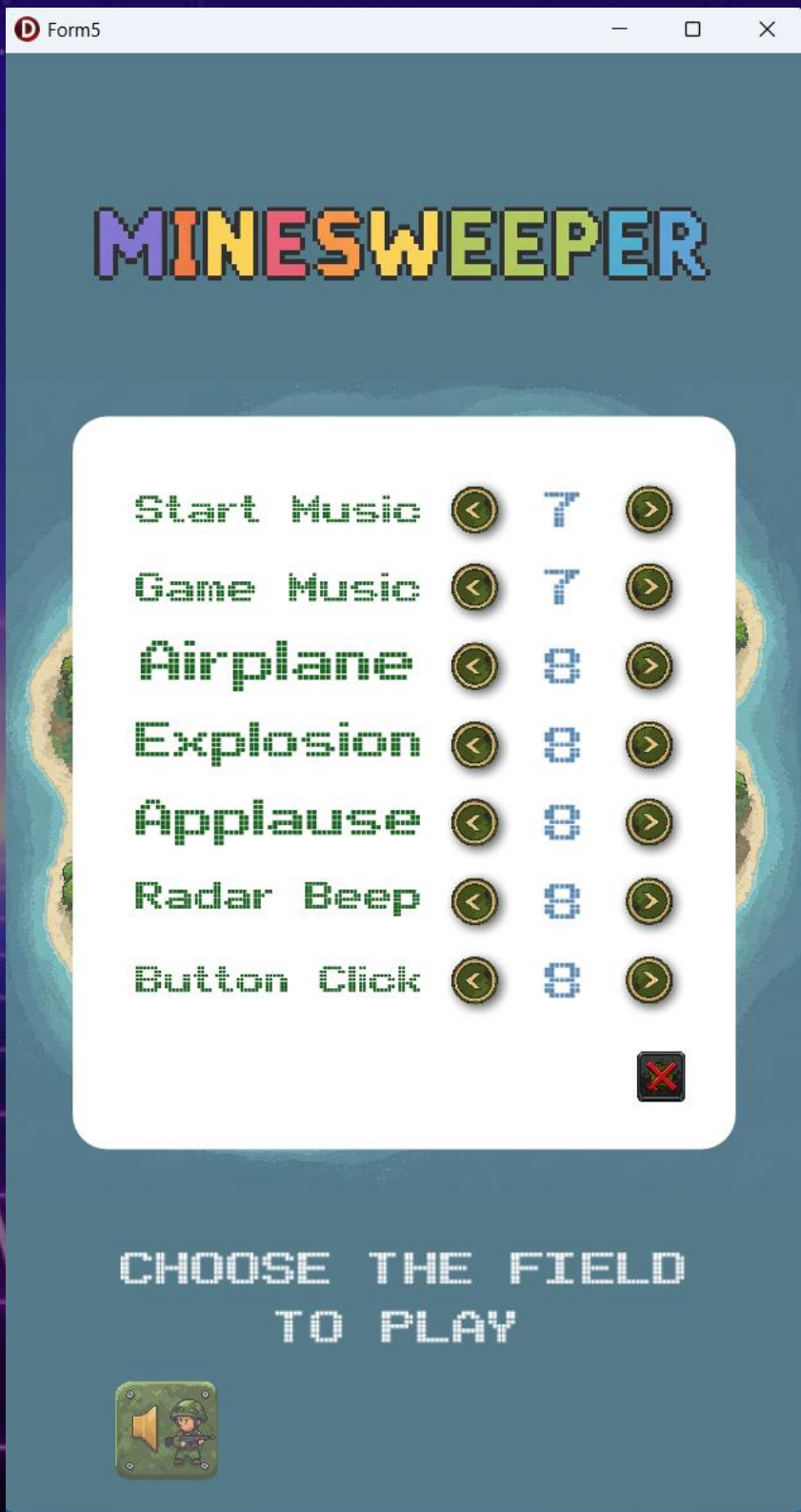

Códigos

Campo de seleção

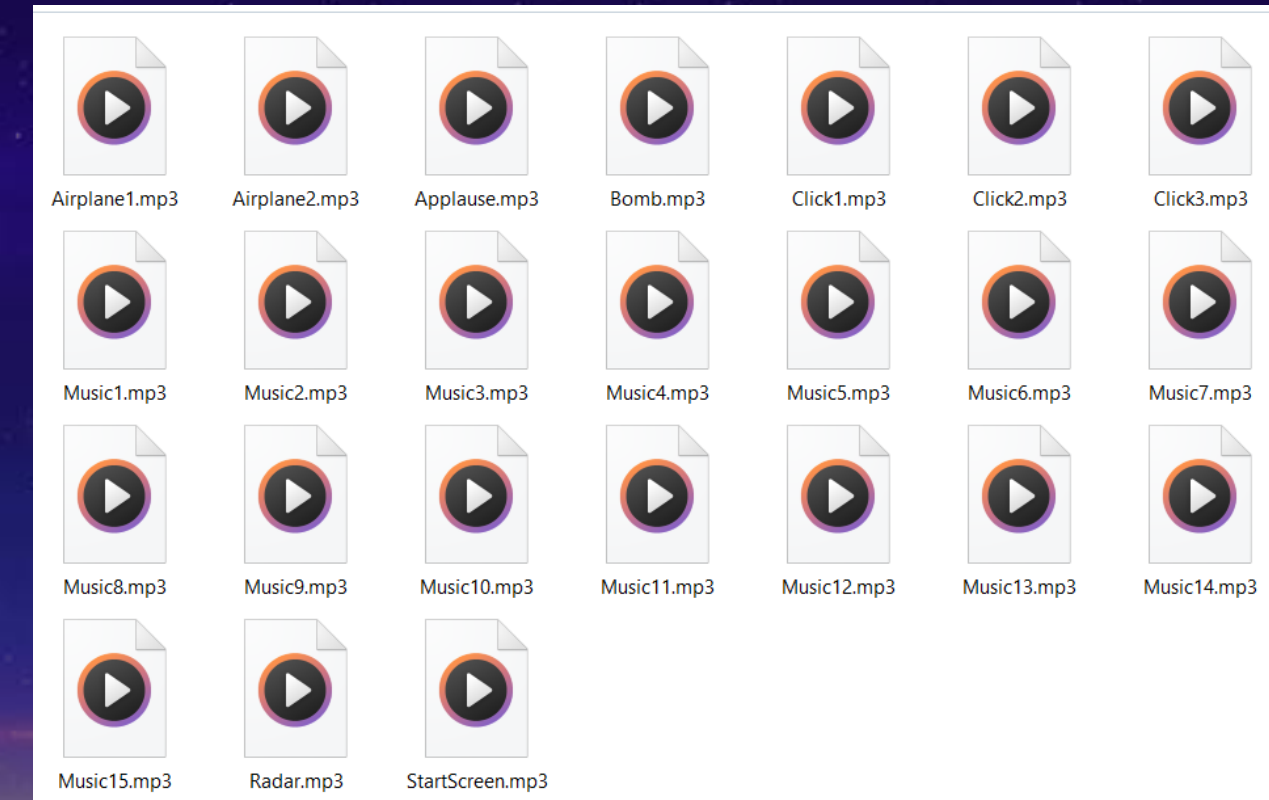
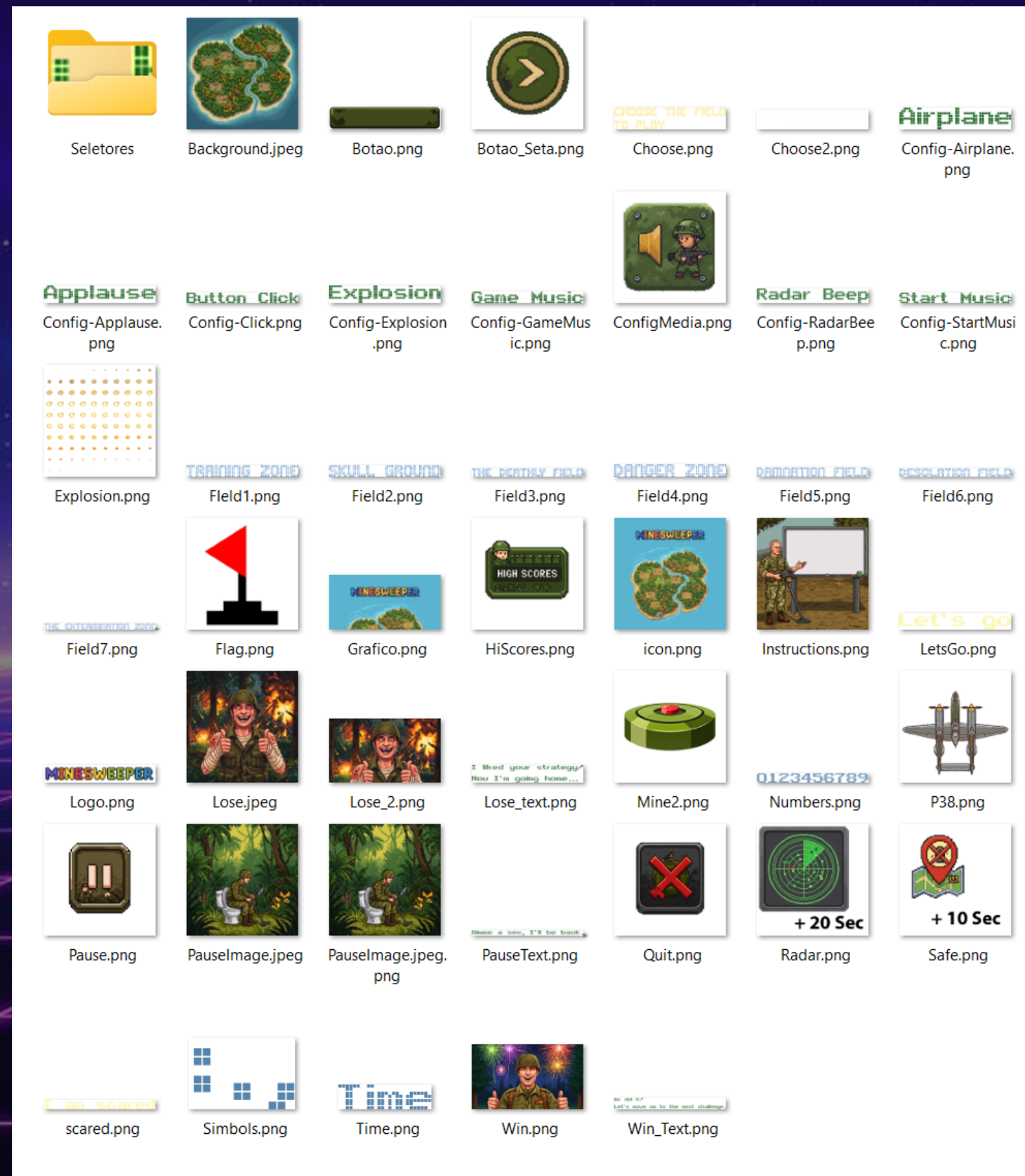


Very Easy	Very Small	Easy	Small	Medium	Hard	Large
01-VeryEasy.png	01-VerySmall.png	02-Easy.png	02-Small.png	03-Medium.png	04-.png	04-Large.png
Very Hard	Very Large					
05-VeryHard.png	05-VeryLarge.png					

Controle de volume



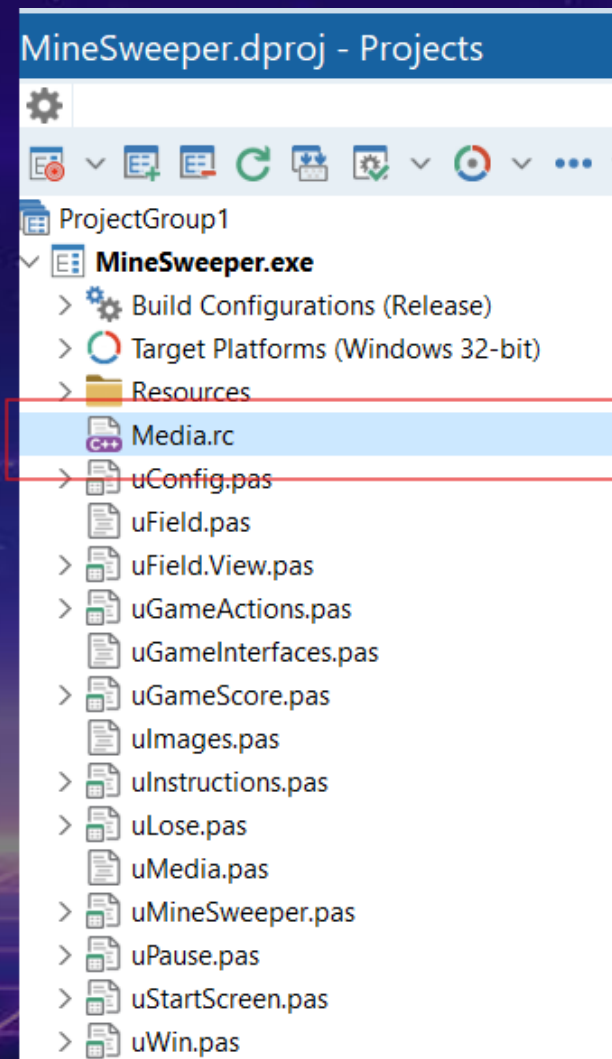
Imagens e trilha sonora - 1



Imagens e trilha sonora - 2

```
uMineSweeper | uSelector | Media.rc | uStartScreen | uInstruio

1 Explosion RCDATA "Imagens\\Explosion.png"
- Flag RCDATA "Imagens\\Flag.png"
- Background RCDATA "Imagens\\Background.jpeg"
- Mine RCDATA "Imagens\\Mine2.png"
- Easy RCDATA "Imagens\\Easy.png"
- Hard RCDATA "Imagens\\Hard.png"
- Large RCDATA "Imagens\\Large.png"
- Medium RCDATA "Imagens\\Medium.png"
- Small RCDATA "Imagens\\Small.png"
10 VeryEasy RCDATA "Imagens\\VeryEasy.png"
- VeryHard RCDATA "Imagens\\VeryHard.png"
- VeryLarge RCDATA "Imagens\\VeryLarge.png"
- VerySmall RCDATA "Imagens\\VerySmall.png"
- Numbers RCDATA "Imagens\\Numbers.png"
- Simbols RCDATA "Imagens\\Simbols.png"
- P38 RCDATA "Imagens\\P38.png"
- Airplane1 RCDATA "Sounds\\Airplane1.mp3"
- Airplane2 RCDATA "Sounds\\Airplane2.mp3"
- bomb RCDATA "Sounds\\bomb.mp3"
20 Click1 RCDATA "Sounds\\Click1.mp3"
- Click2 RCDATA "Sounds\\Click2.mp3"
- Click3 RCDATA "Sounds\\Click3.mp3"
- StartScreen RCDATA "Sounds\\StartScreen.mp3"
- Applause RCDATA "Sounds\\Applause.mp3"
- Radar RCDATA "Sounds\\Radar.mp3"
- Music1 RCDATA "Sounds\\Music1.mp3"
- Music2 RCDATA "Sounds\\Music2.mp3"
- Music3 RCDATA "Sounds\\Music3.mp3"
- Music4 RCDATA "Sounds\\Music4.mp3"
30 Music5 RCDATA "Sounds\\Music5.mp3"
- Music6 RCDATA "Sounds\\Music6.mp3"
- Music7 RCDATA "Sounds\\Music7.mp3"
- Music8 RCDATA "Sounds\\Music8.mp3"
- Music9 RCDATA "Sounds\\Music9.mp3"
- Music10 RCDATA "Sounds\\Music10.mp3"
- Music11 RCDATA "Sounds\\Music11.mp3"
- Music12 RCDATA "Sounds\\Music12.mp3"
- Music13 RCDATA "Sounds\\Music13.mp3"
- Music14 RCDATA "Sounds\\Music14.mp3"
40 Music15 RCDATA "Sounds\\Music15.mp3"
-
```



```
constructor TMedia.Create(aOwner: TComponent);

Procedure TestMediaFiles;

    Procedure TestResource(aResName : String);
    var
        LResStream : TResourceStream;
        LFile       : String;
    begin
        LFile := TPath.Combine(FPath, aResName+'.mp3');
        If Not FileExists(LFile) Then
            Begin
                LResStream := TResourceStream.Create(HInstance, aResName, RT_RCDATA);
                LResStream.SaveToFile(LFile);
            End;
        End;

    Begin
        FPath := TPath.Combine(TPath.GetDocumentsPath, 'GameAudio');
        If Not DirectoryExists(FPath) Then
            CreateDir(FPath);
        TestResource('AirPlane1');
        TestResource('AirPlane2');
```


Case 2 - Arkanoid

<https://play.google.com/store/apps/details?id=br.com.imperium.Arkanoid>

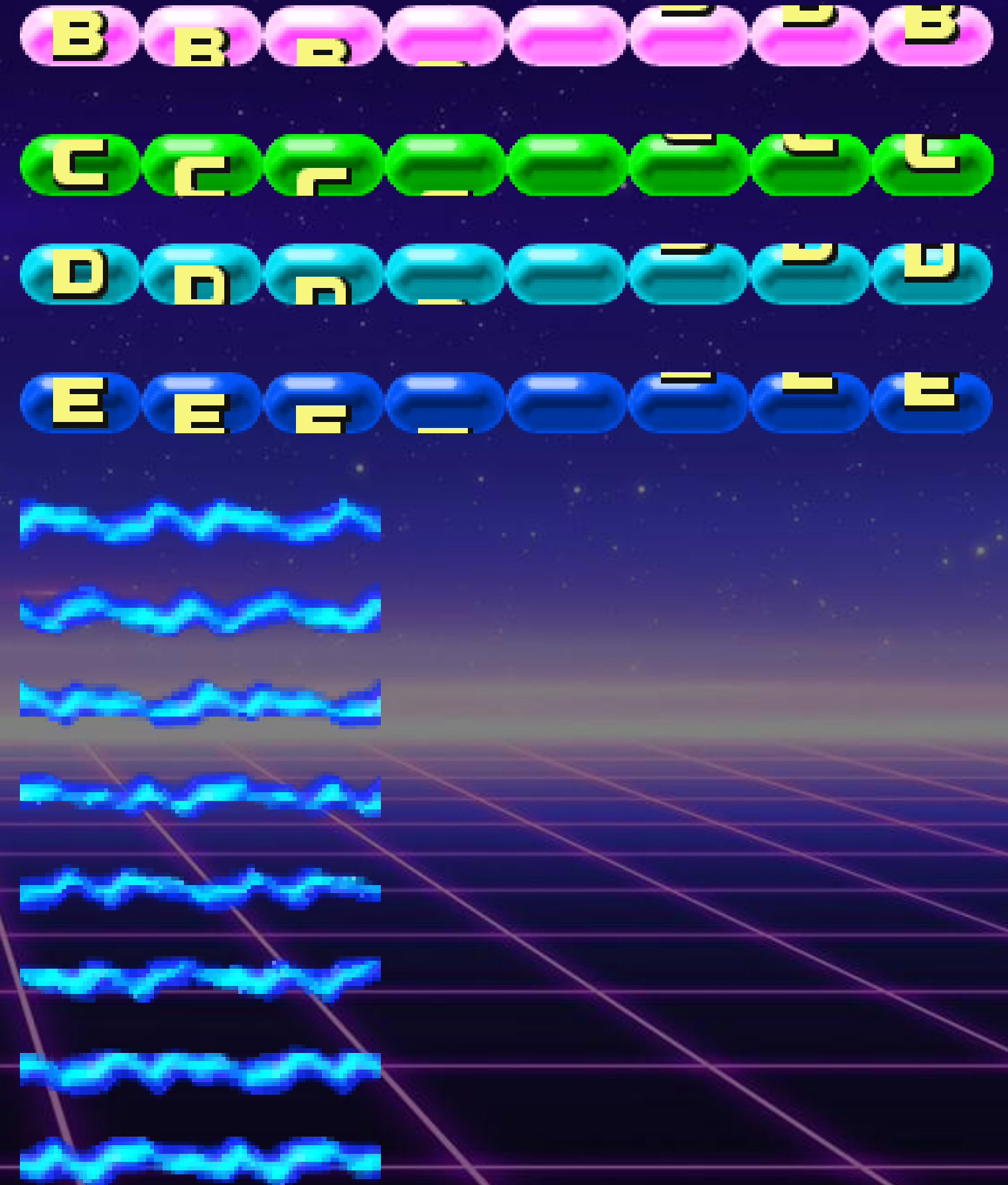
Surgiu como uma POC em um grupo do WhatsApp.



TBitmapListAnimation

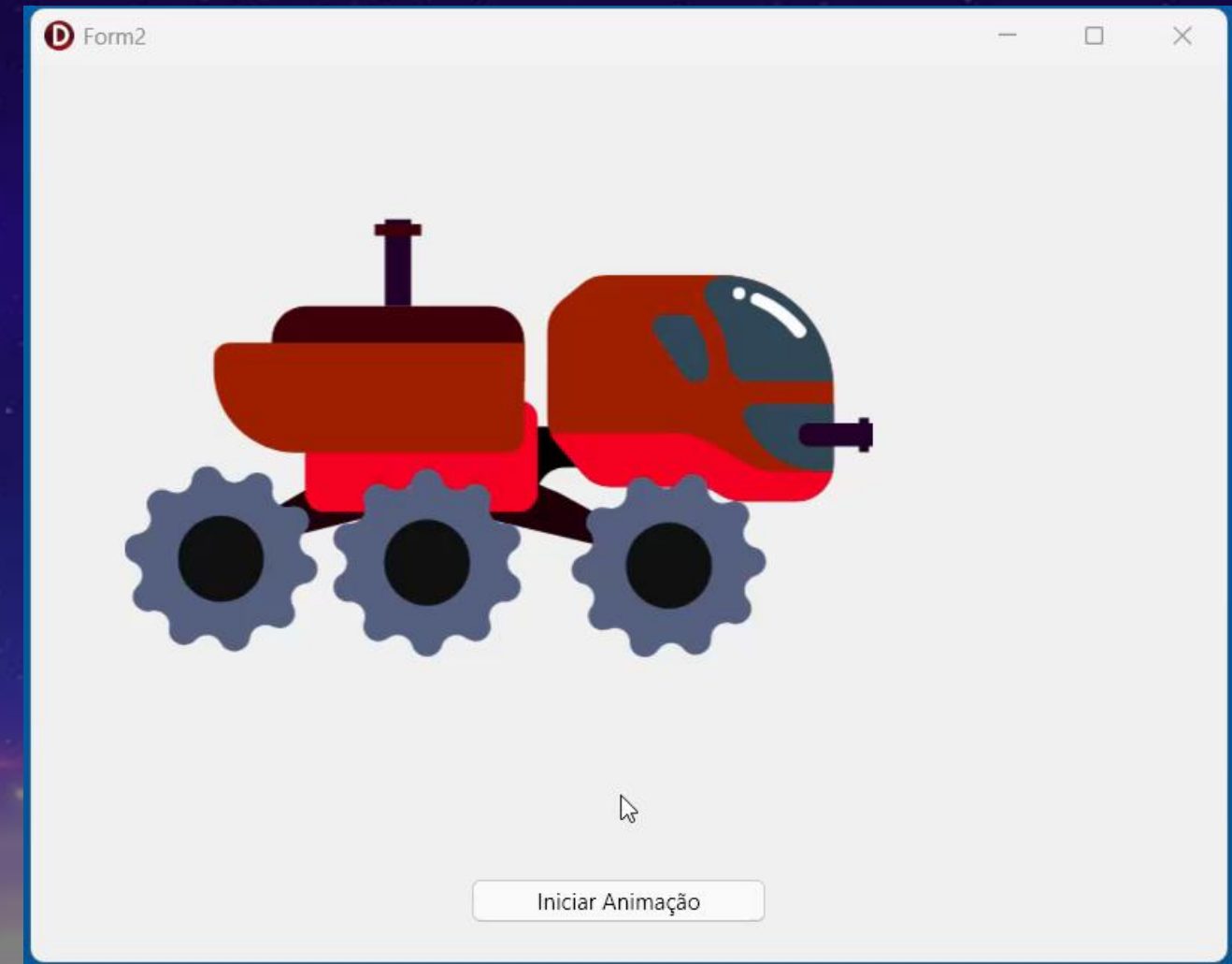
Animação de sprites

```
. Fill.Bitmap.WrapMode := TWrapMode.Tile;  
72 FAni := TBitmapListAnimation.Create(Self);  
. FAni.Parent := Self;  
. FAni.PropertyName := 'Fill.Bitmap.Bitmap';  
- FAni.AnimationCount := 8;  
. FAni.AnimationRowCount := 1;  
. FAni.Loop := True;  
. FAni.Duration := 0.8;  
. case aTipo of  
80 tiMaisVida : ResName := 'V';  
. tiGrande : ResName := 'G';  
. tiPequeno : ResName := 'P';  
. tiCampoForca : ResName := 'C';  
. tiMaisBolas : ResName := 'B';  
- tiTiro : ResName := 'T';  
. tiTiraVida : ResName := 'M';  
. tiSuperBola : ResName := 'S';  
. end;  
- FAni.AnimationBitmap := TResource.GetBitmap(ResName);
```



Usando SKIA para animar um SVG

```
type
-
-   TRover = Class(TSKSvg)
-       Private
-           FAniCab : TFloatAnimation;
-           FAngle1 : Single;
-           FAngle2 : Single;
-           FAngle3 : Single;
-           FYWheel1 : Single;
-           FYWheel2 : Single;
-           FYWheel3 : Single;
-           FYCabine : Single;
-           procedure SetYCabine(const Value: Single);
-       Public
-           Constructor Create(aOwner : TComponent); Override;
-           Procedure Start;
-           Procedure Stop;
-           Property YCabine : Single Read FYCabine Write SetYCabine;
-           Property YWheel1 : Single Read FYWheel1 Write FYWheel1;
-           Property YWheel2 : Single Read FYWheel2 Write FYWheel2;
-           Property YWheel3 : Single Read FYWheel3 Write FYWheel3;
-       End;
```



```
208 procedure TRover.SetYCabine(const Value: Single);
-   begin
210     FYCabine := Value;
-     Svg.DOM.FindNodeById('Cabine').TrySetAttribute('transform', 'translate(0 '+FYCabine.ToString.Replace(',', '.')+')');
-     FAngle1 := FAngle1 + 2;
-     FAngle2 := FAngle2 + 2;
-     FAngle3 := FAngle3 + 2;
-     Svg.DOM.FindNodeById('Wheel1').TrySetAttribute('transform', 'translate(-67.7611 '+(-310.425+FYWheel1).ToString.Replace(',', '.')+') rotate('+(FAngle1).ToString+', 402, 520)');
-     Svg.DOM.FindNodeById('Wheel2').TrySetAttribute('transform', 'translate(-67.7611 '+(-310.425+FYWheel2).ToString.Replace(',', '.')+') rotate('+(FAngle2).ToString+', 255, 520)');
-     Svg.DOM.FindNodeById('Wheel3').TrySetAttribute('transform', 'translate(-67.7611 '+(-310.425+FYWheel3).ToString.Replace(',', '.')+') rotate('+(FAngle3).ToString+', 129, 520)');
-     if FYWheel1 <> 0 then Svg.DOM.FindNodeById('Axle1').TrySetAttribute('transform', 'translate(-67.7611 -310.425) rotate('+(FYWheel1).ToString.Replace(',', '.')+ ', 315, 500)');
-     if FYWheel3 <> 0 then Svg.DOM.FindNodeById('Axle3').TrySetAttribute('transform', 'translate(-67.7611 -310.425) rotate('+(-FYWheel3).ToString.Replace(',', '.')+ ', 199, 450)');
220     Redraw;
-   end;
```



```
constructor TROver.Create(aOwner: TComponent);
begin
  inherited;
  SVG.Source :=
    '<?xml version="1.0" encoding="UTF-8"?>'+
    '<svg xmlns="http://www.w3.org/2000/svg" id="Camada_1" data-name="Camada 1" viewBox="0 0 456.9382 267.2698" transform="rotate(0)">'+
    '  <path d="M367.547,443.89v-6.91H310.023v51.633h10.399V477.339c0-8.484,7.836-15.367,17.508-15.367h29.613V443.89Zm17.195-126.977" transform="translate(-67.7611 -310.425)" />'+
    '  <path id="Axle1" transform="translate(-67.7611 -310.425) rotate(0, 315, 500)" '+
    '    d="M382.494,515.0061l3.4019-1.7172c2.0673-1.0435,2.1051-5.8376.055-6.9691l-51.2827-28.3042c-.1439-.0793-7.8566-3.3443-13.7791-5.8235a25.2621,25.2621,0,0,0-13.8053-1.5962L258.0022,478.65'+
    '    c-2.7521.03-3.025,6.64-.3007,7.2829L381.5266,515.13A1.4,1.4,0,0,0,382.494,515.0061Z" style="fill:#1f0009" />'+
    '  <path id="Axle2" transform="translate(-67.7611 -310.425) rotate(0, 0, 0)" '+
    '    d="M264.9982,482.3091c-20.824,46.3569,3.2534-7.8457-10.7376,24.966a4.5786,4.5786,0,0,1-8.1065.6145l-12.6524-24.9132" style="fill:#1f0009" />'+
    '  <path id="Axle3" transform="translate(-67.7611 -310.425) rotate(0, 199, 450)" '+
    '    d="M132.596,512.9182l-2.3911-1.2764c-2.1953-1.1718-2.2355-6.5557-.0585-7.8263l47.6884-27.8344a1.4784,1.4784,0,0,1,.7663-.2132l68.3025.7838c2.9224.0335,3.2123,7.4573.3193,8.1786l-113.6,28.327A1.4141'+
    '    ,1.4141,0,0,1,132.596,512.9182Z" style="fill:#1f0009" />'+
    '  <g id="Cabine" transform="translate(0 0)">'+
    '    <path d="M326.188,378.511V432.14a22.6432,22.6432,0,0,0,5.476,14.771l16.332,18.972a22.6374,22.6374,0,0,0,17.156,7.867H417.18a22.6878,22.6878,0,0,1,13.172,4.219,25.2742,25.2742,0,0,0,14.703,4.715h33.281a2'+
    '      2.9282,22.9282,0,0,0,22.926-22.926V413.902a69.3753,69.3753,0,0,0-69.375-69.375H362.25a24.1237,24.1237,0,0,0-15.359,5.523l-11.95,9.871a24.0983,24.0983,0,0,0-8.753,18.59Zm0,0" '+
    '      transform="translate(-67.7611 -310.425)" style="fill:#9f1f00" />'+
    '    <path d="M501.105,409.456a69.3572,69.3572,0,0,0-67.394-64.886c-1.629.336-12.133,2.84-11.809,11.968a11.9354,11.9354,0,0,0,2.528,6.645,64.8432,64.8432,0,0,1,12.461,33.793,13.15,13.15,0,0,0,13.043,12.48Zm'+
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    '      7.18a22.6878,22.6878,0,0,1,13.172,4.219,25.2742,25.2742,0,0,0,14.703,4.715h33.281a22.9324,22.9324,0,0,0,22.391-18.031Zm0,0" transform="translate(-67.7611 -310.425)" style="fill:#f40022" />'+
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    '    .93Zm0,0" transform="translate(-67.7611 -310.425)" style="fill:#f40022" />'+
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    '  </g>'+
    '</svg>';
```


MineSweeper

<https://github.com/ImperiumDelphi/minesweeper>

Arkanoid

<https://github.com/ImperiumDelphi/Arkanoid>

Rover MoonPatrol

https://github.com/ImperiumDelphi/Rover_MoonPatrol

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